Chapter 1
ANALOG CONCEPTS

CHAPTER OUTLINE

1-1 Devices and Quantities

1-2 Signals

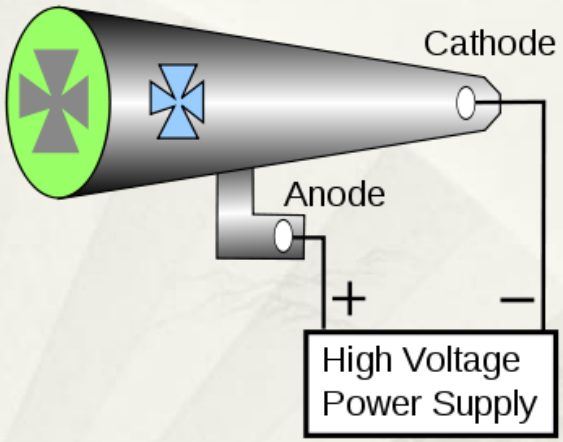
1-3 Troubleshooting Analog Circuits

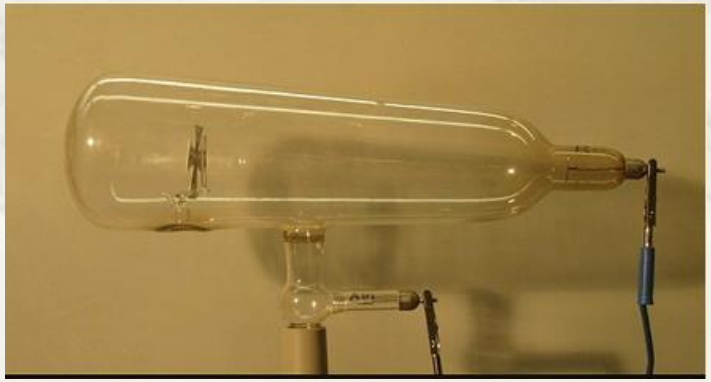
SCIENCE HIGHLIGHT

- Crookes Experiment and Electron



William Crookes
(1832～1919)

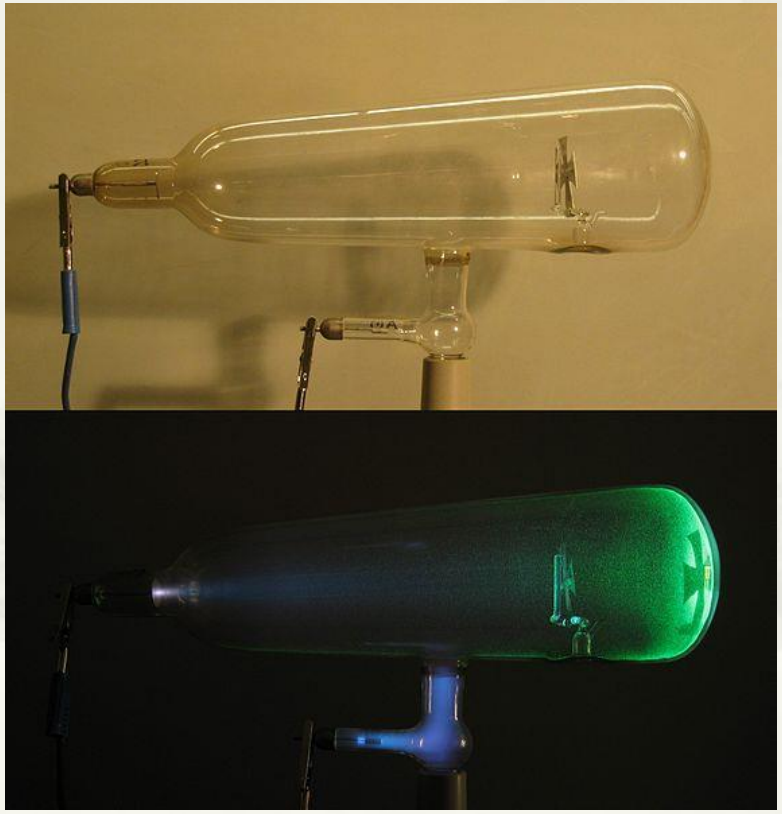




The Crookes Tube

SCIENCE HIGHLIGHT

- Crookes Experiment and Electron



The Crookes Tube



Cathode Ray Tube (CRT)

1-1 DEVICES AND QUANTITIES

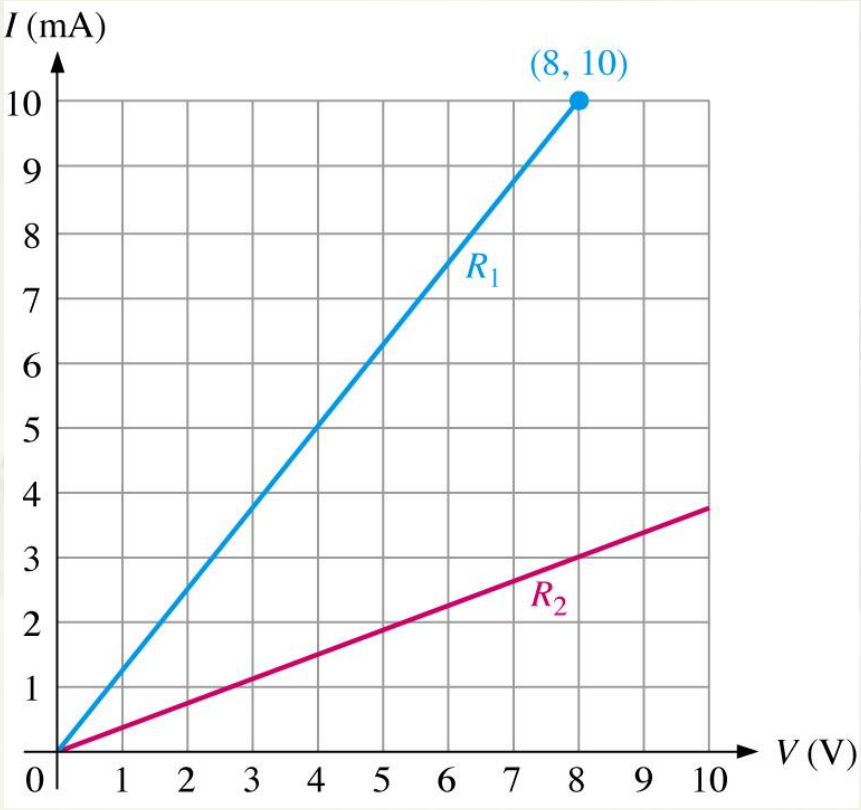
- Electronic Devices
- Linear Equations
- AC Resistance
- Conventional Current Versus Electronic Flow

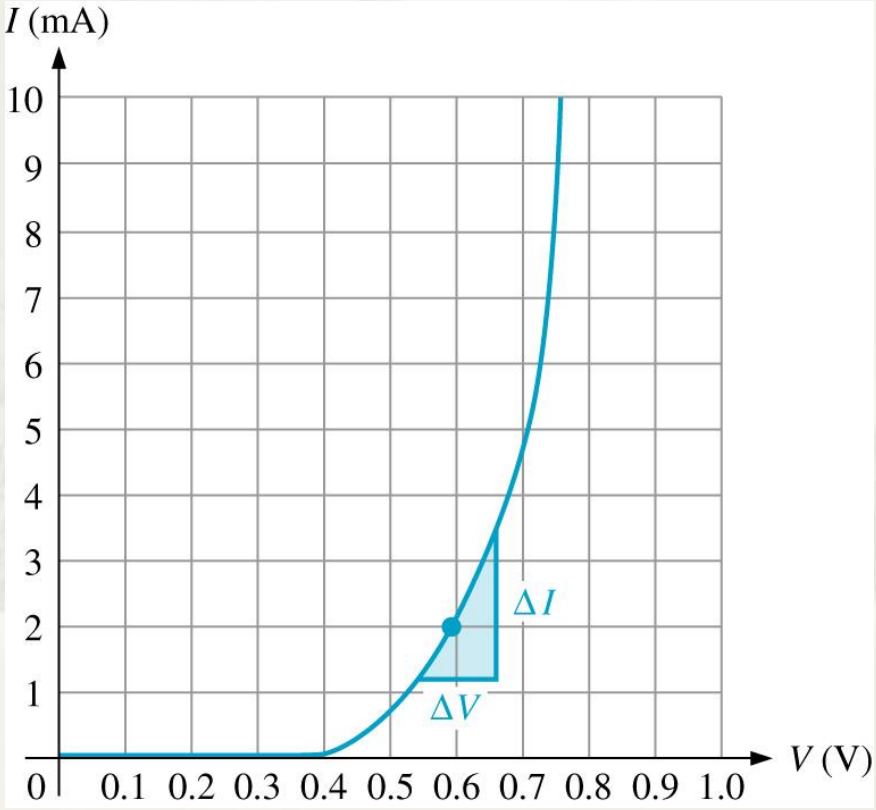
Electronic Devices

- **Electronics Progressing**
  - Spark-gap ⟶ Vacuum Tube ⟶ Transistor ⟶ IC
- **Integrated Circuit**
  - Analog IC
  - Digital IC
- **Components and Devices**
  - Passive Components
  - Active Components

Electronic Devices

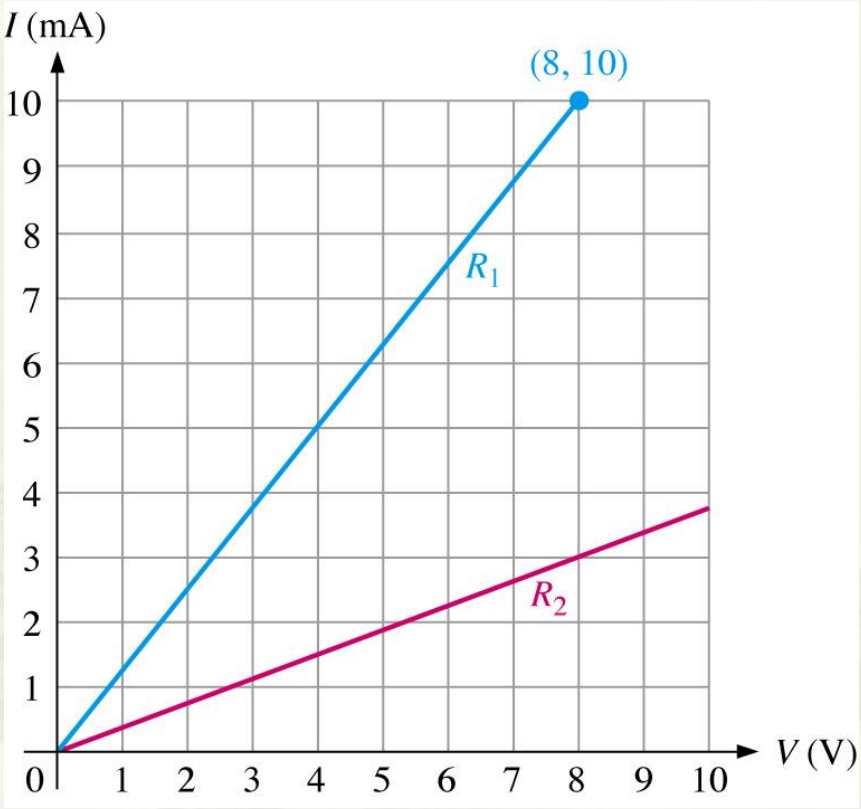
- **Describing Basic Components using *IV* Curves**





Linear Equations

- **Linear Equation**
  - $y = mx + b$
  - $y = mx$
- **Ohm's Law**
  - $I = \frac{V}{R} = GV$
- **Linear Component**
  - A linear component is one in which an increase in current is proportional to the applied voltage as given by Ohm's law.



IV characteristic curves for two resistors

Example 1-1

Problem

Figure 1–1 shows the characteristic curves for two resistors. What are the conductance and resistance of R_1?

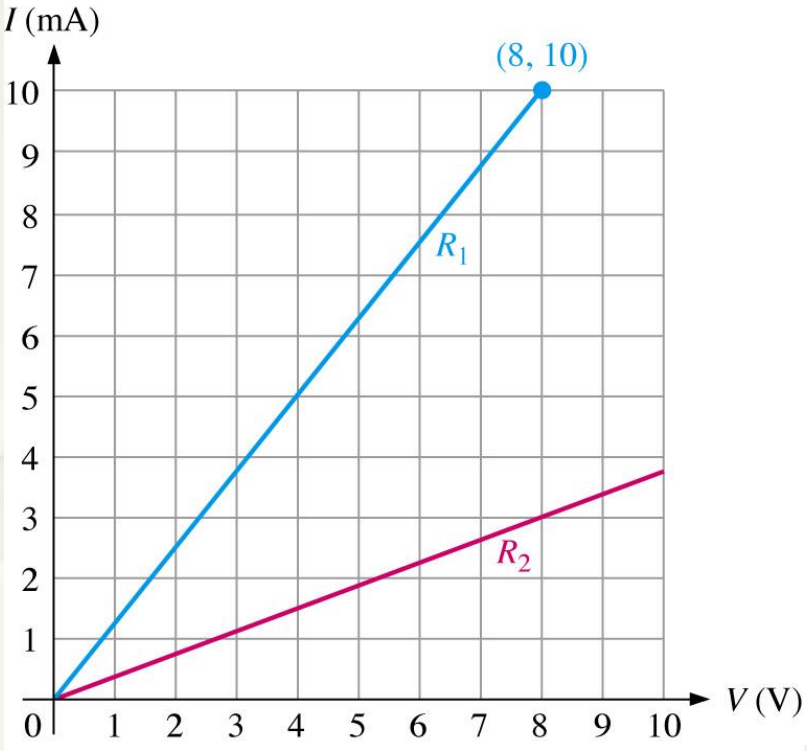


Figure 1-1 *IV* characteristic curves for two resistors.

Solution

Find the conductance, G_1, by measuring the slope of the *IV* characteristic curve for R_1. The slope is the change in the y variable (written Δy) divided by the corresponding change in the x variable (written Δx).

$$\text{slope} = \frac{\Delta y}{\Delta x}$$

For a straight line, the slope is constant, so you can use any two points to determine the conductance.

Example 1-1

Problem

Figure 1–1 shows the characteristic curves for two resistors. What are the conductance and resistance of R_1?

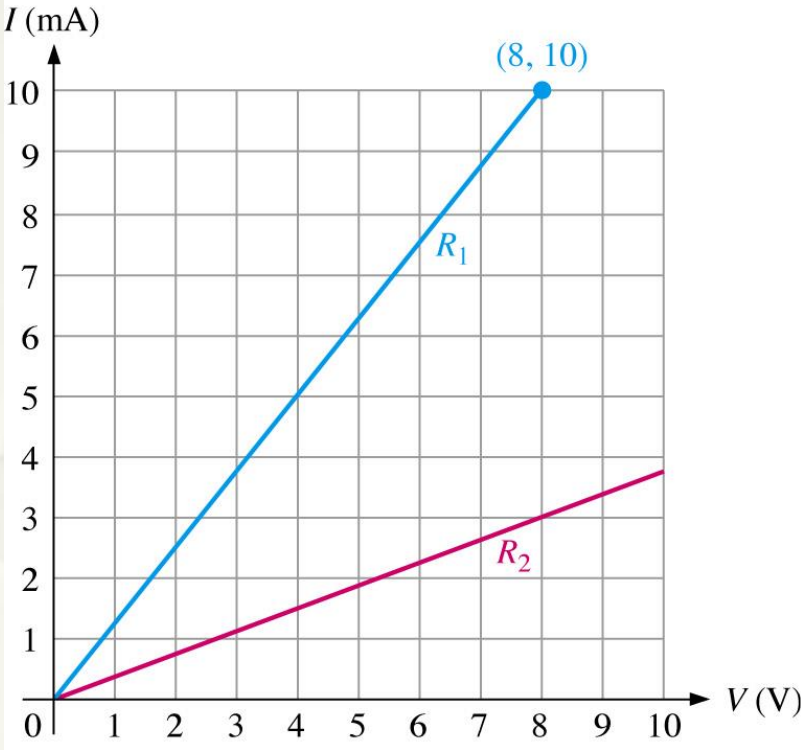


Figure 1-1 *IV* characteristic curves for two resistors.

Solution

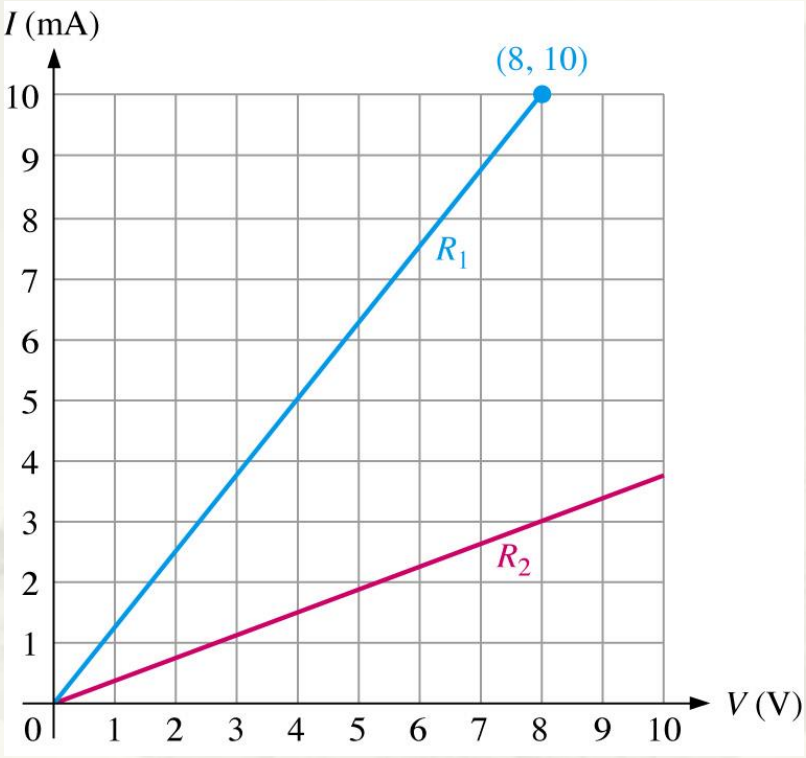
Choosing the point (x = 8 V, y = 10 mA) from Figure I-I as one point and the origin (x = 0 V, y = 0 mA) as the other point, you can find the slope and, therefore, the conductance as.

$$G_1 = \frac{10\text{ mA} - 0\text{ mA}}{8.0\text{ V} - 0\text{ V}} = \mathbf{1.25\text{ mS}}$$

The resistance is the reciprocal of the conductance.

$$R_1 = \frac{1}{G_1} = \frac{1}{1.25\text{ mS}} = \mathbf{0.8\text{ k}\Omega}$$

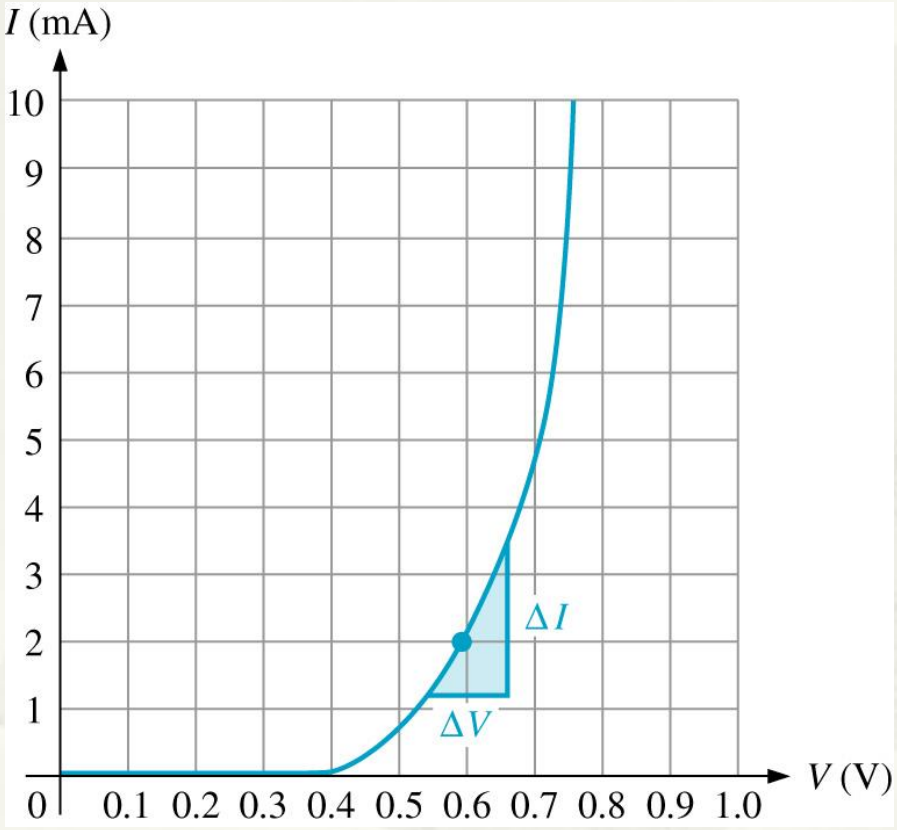
AC Resistance



IV characteristic curves for two resistors

- **DC Resistance**
  - The ratio of voltage at some point to the corresponding current at that point is referred as the *dc resistance*.

AC Resistance



An *IV* characteristic curve for a diode

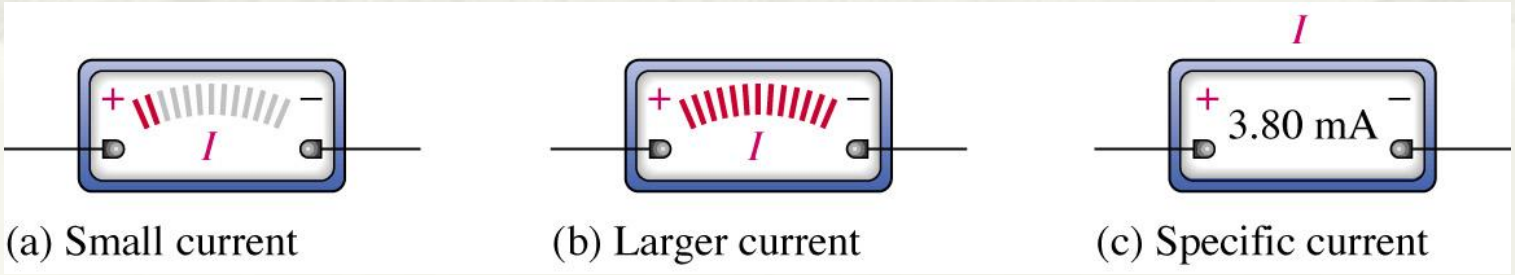
- **AC Resistance (Dynamic Resistance)**
  - The ratio of a small change in voltage divided by the corresponding small change in current is defined as the *ac resistance* of an analog device.

$$r_{ac} = \frac{\Delta V}{\Delta I} = \frac{0.66V - 0.54V}{3.4mA - 1.2mA}$$

$$= 54.5\Omega$$

Conventional Current Vs. Electron Flow

- **Conventional current**
  - It assumes that currents is out of positive terminal of a voltage source, through the circuit, and into the negative terminal of the source.
- **Electron flow**
  - It is known that in solid metallic conductors, the moving charge is actually the negatively charged electrons , which move from the negative to the positive point.



Special meter symbols for showing current

1-2 SIGNALS

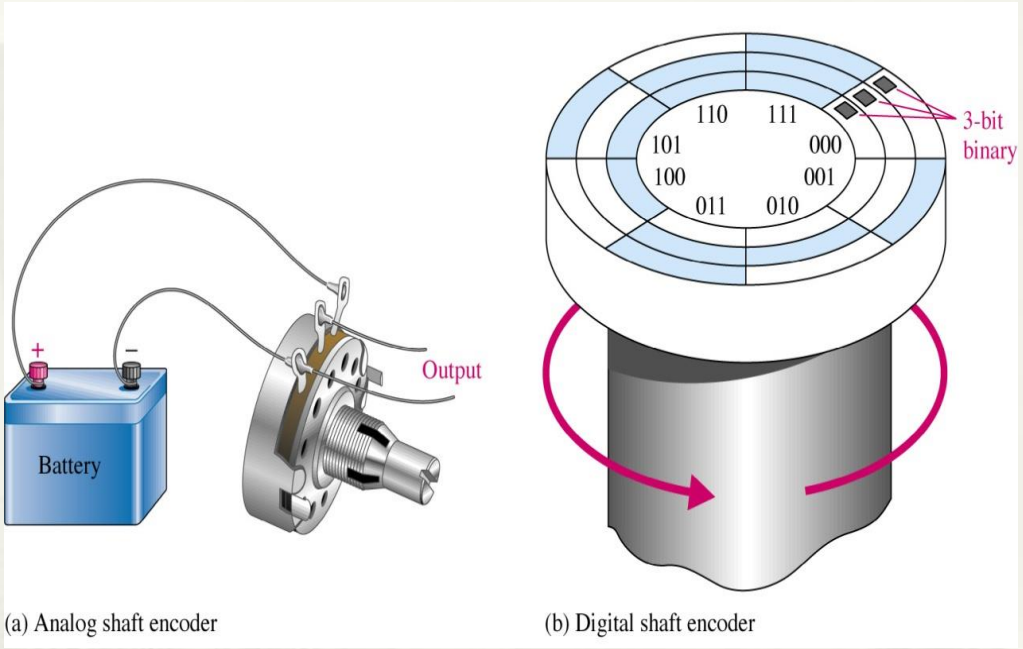
- Analog and Digital Signals
- Periodic Signals
- Time-Domain Signals
- Frequency-Domain Signals

Signal

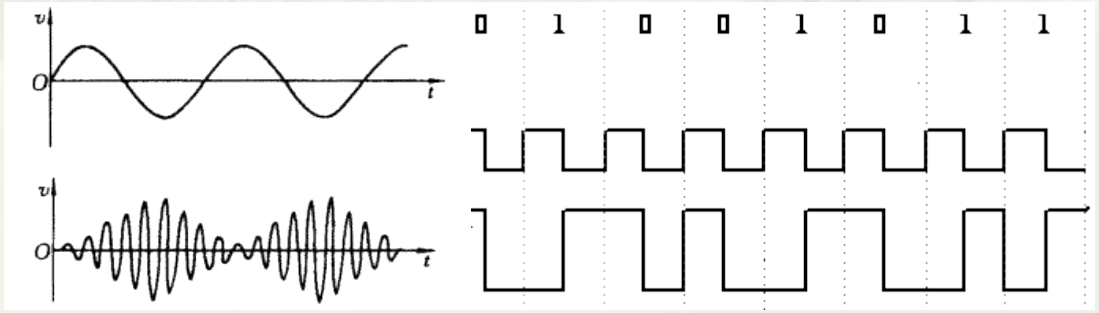
- **Signal**
  - An electrical voltage or current that contains information
  - The information is carried by electrical waves, either in a ***conductor*** **(conduction)** or as an ***electromagnetic field*** **(radiation).**

Analog and Digital Signals

- **Signals processing**
  - Modify the shape
  - Amplify the power level
- **Signals classifying**
  - Continuous ⟶ Analog Signal
  - Discrete (time and values) ⟶ Digital Signal



Analog and digital shaft encoders

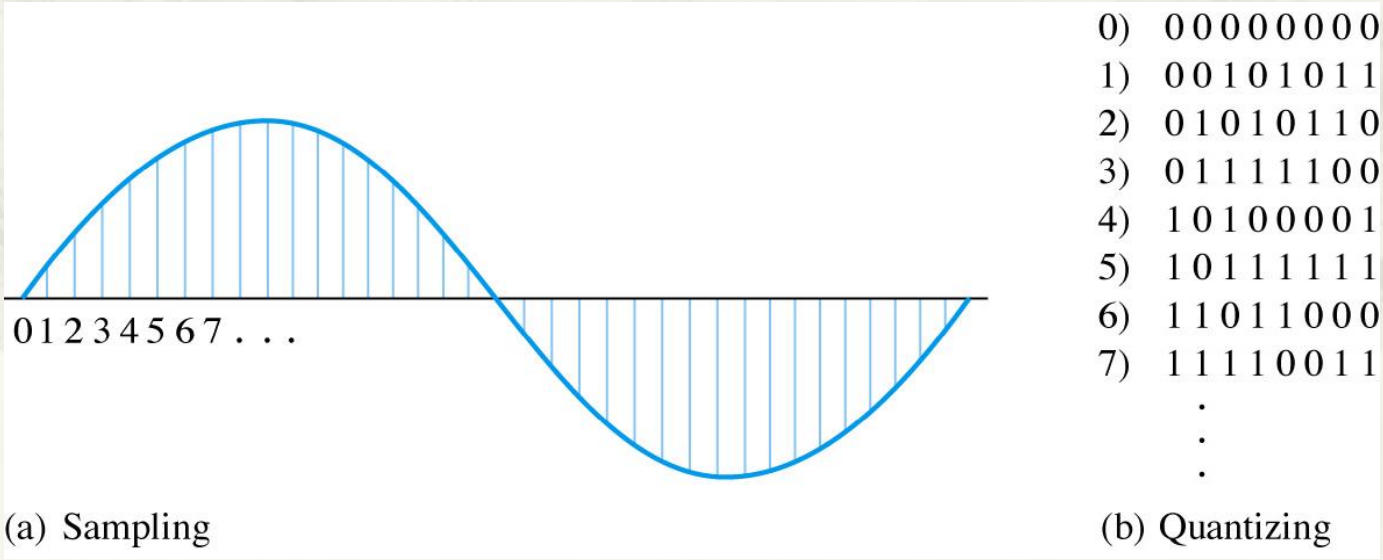


Analog and Digital Circuit

- **Circuit classification**
  - Analog circuit : Processing analog signal
    - ✓ Simple, high speed, low cost, readily simulate nature phenomena
  - Digital circuit : Processing Digital signal
    - ✓ High noise immunity, no drift, high speed, ability to perform various mathematical calculations
- **Transducer**
  - A device that converts energy from one form to another.

Analog – Digital Conversion

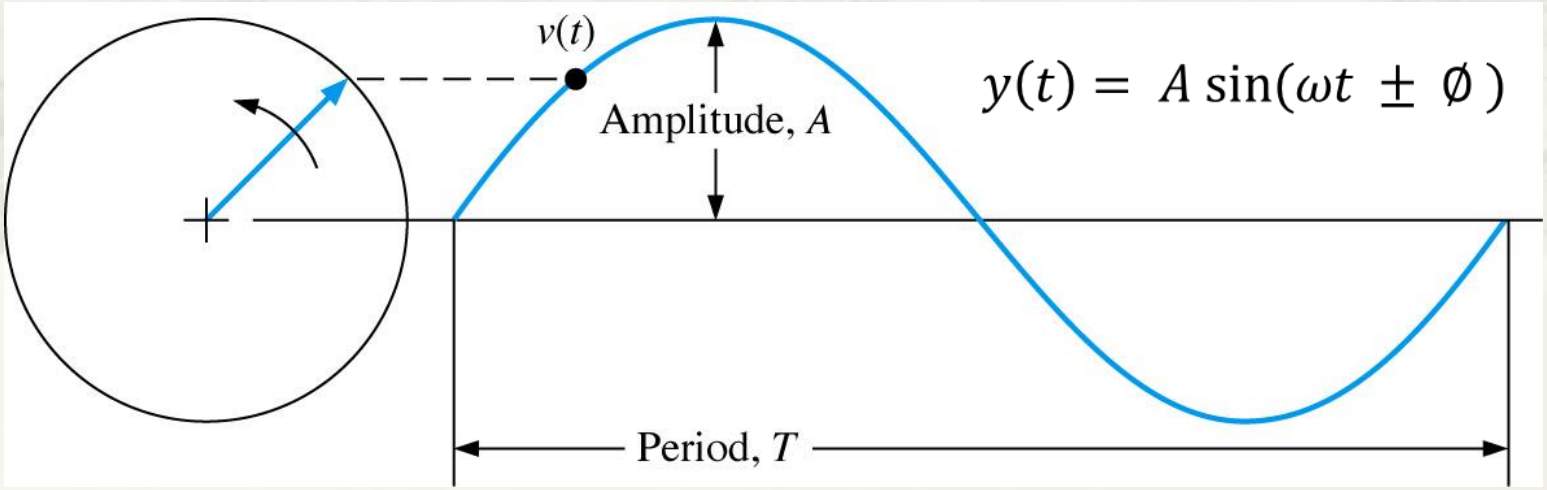
- **Sampling**
  - The process of breaking an analog waveform into time slices that approximate the original wave.
- **Quantizing**
  - Assigning a numeric value for the time slice.



(a) Sampling (b) Quantizing

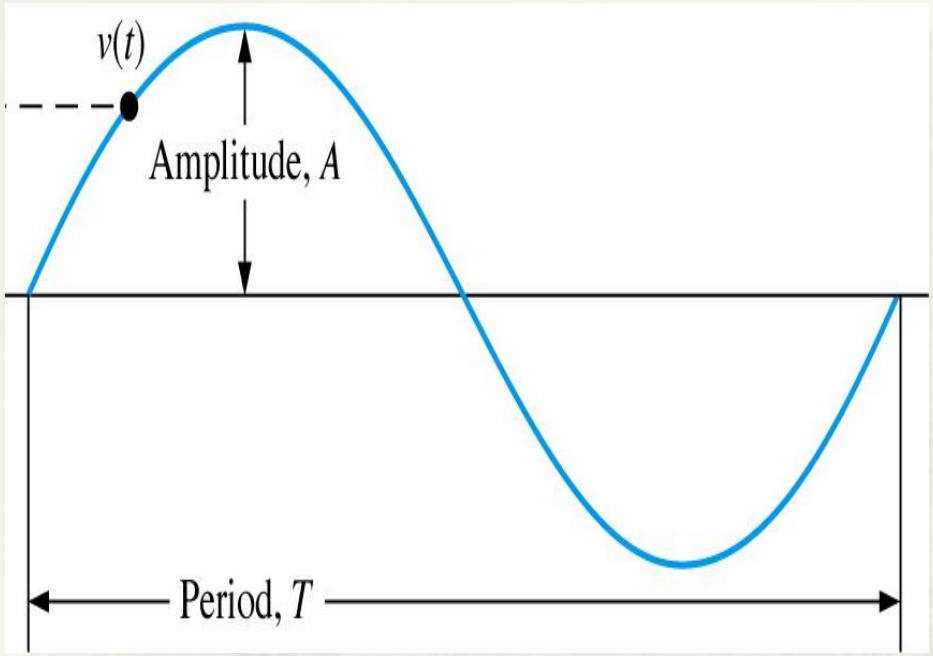
Periodic Signals

- **Periodic signals**
  - ***Sinusoidal wave***
  - Nonsinusoidal waves
    - ✓ *Square, rectangular, triangle, sawtooth, etc.*
  - Generated by ***Oscillators***



Generation of a sinusoidal waveform from the projection of a rotating vector

Sinusoidal Wave Parameters

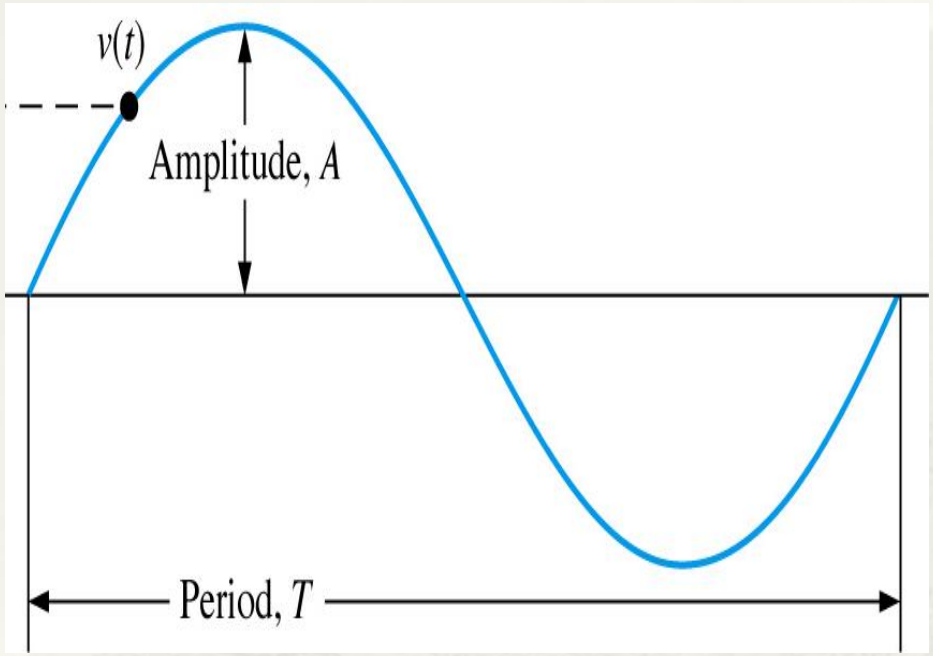


$$y(t) = A\sin(\omega t \pm \emptyset)$$

ω -- angular frequency

- **Frequency**
  - $f(Hz) = \dfrac{\omega(\text{rad/s})}{2\pi(\text{rad/cycle})}$
- **Period**
  - $T = \dfrac{1}{f}$
- **Instantaneous Value**
  - $v(t) = V_p \sin(\omega t \pm \emptyset)$
- **Peak Value**
  - V_p
  - $V_{pp} = 2V_p$

Sinusoidal Wave Parameters



$$y(t) = A\sin(\omega t \pm \emptyset)$$

rms -- root mean square

- **Average Value**
  - $V_{avg} = \frac{\int_0^T |v(t)| \cdot dt}{T}$

    $= \frac{2V_p}{\pi}$ (For sine wave)

    $= 0.637V_p$
- **Effective Value**
  - $V_{rms} = \frac{\sqrt{\int_0^T v(t)^2 \cdot dt}}{T}$

    $= \frac{V_p}{\sqrt{2}} = 0.707V_p$

    (For sine wave)

Example 1-2

Problem

A certain voltage waveform is described by the following equation:

$$v(t) = 15\ \text{V}\ \sin(600t)$$

(a) From this equation, determine the peak voltage and the average voltage. Give the angular frequency in rad/s.

(b) Find the instantaneous voltage at a time of 10 ms.

Example 1-2

Solution

(a) The form of the equation is

$$y(t) = A \sin(\omega t)$$

The peak voltage is the same as the amplitude *(A)*.

$$V_p = \mathbf{15\ V}$$

The average voltage is related to the peak voltage.

$$V_{avg} = 0.637V_p = 0.637(15\ \text{V}) = \mathbf{9.56\ V}$$

The angular frequency, ω, is **600 rad/s**.

(b) The instantaneous voltage at a time of 10 ms is

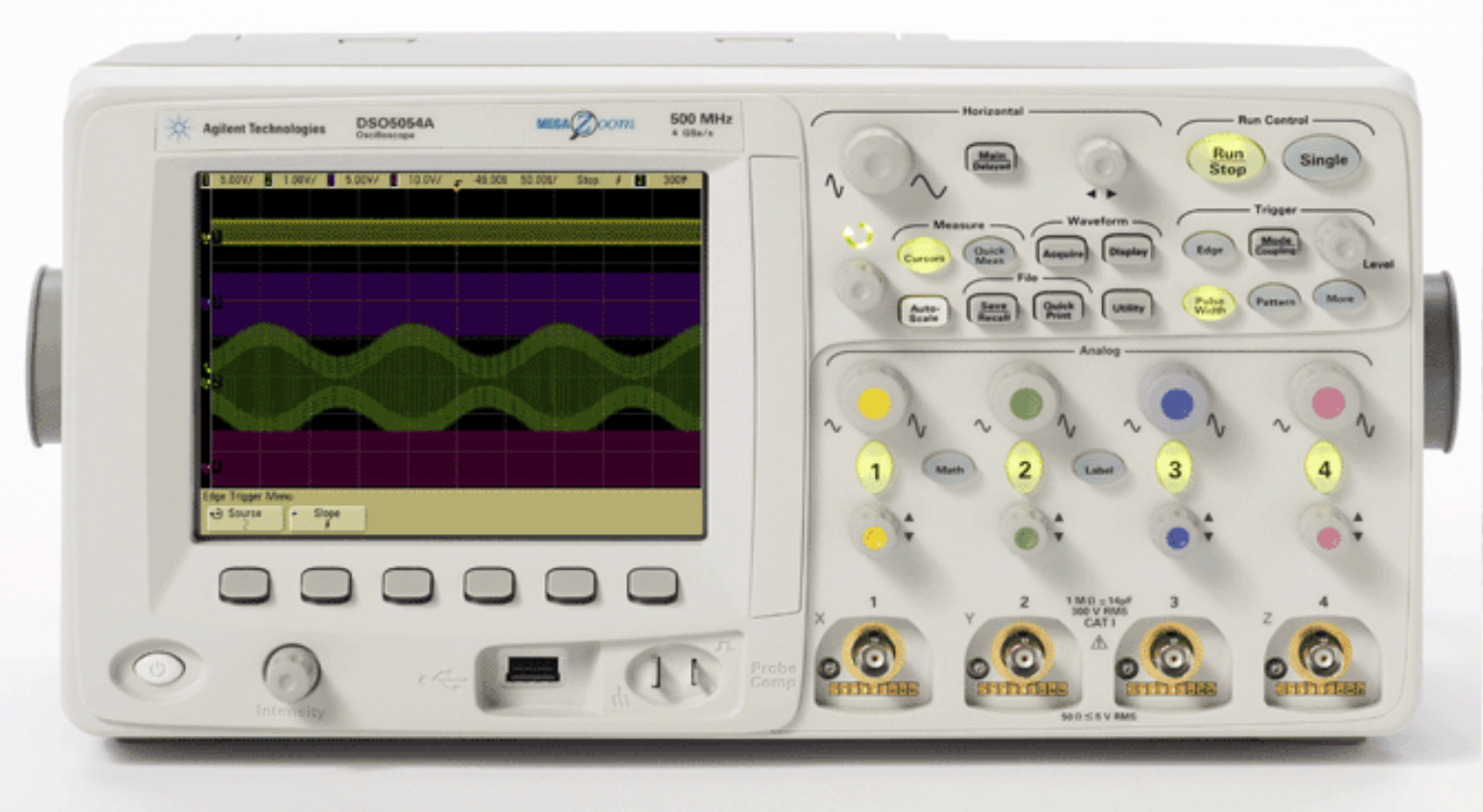
$$v(t) = 15\ \text{V} \sin(600t) = 15\ \text{V} \sin(600\ \text{rad/s})(10\ \text{ms}) = \mathbf{-4.19\ V}$$

Time-Domain Signals

- **Domain**
  - The values assigned to the independent variable are called the domain.
- **Time-domain signals**
  - Signals that have voltage, current, resistance, or other quantity vary as a ***function of time*** are called time-domain signals.
  - Time-domain signals are viewed by ***oscilloscope***.

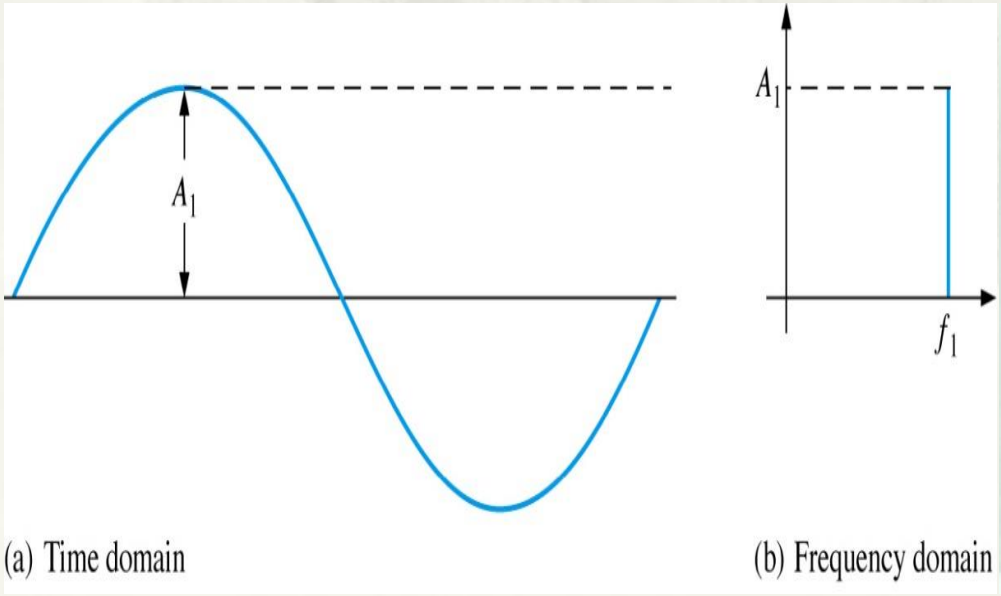
Time-Domain Signals

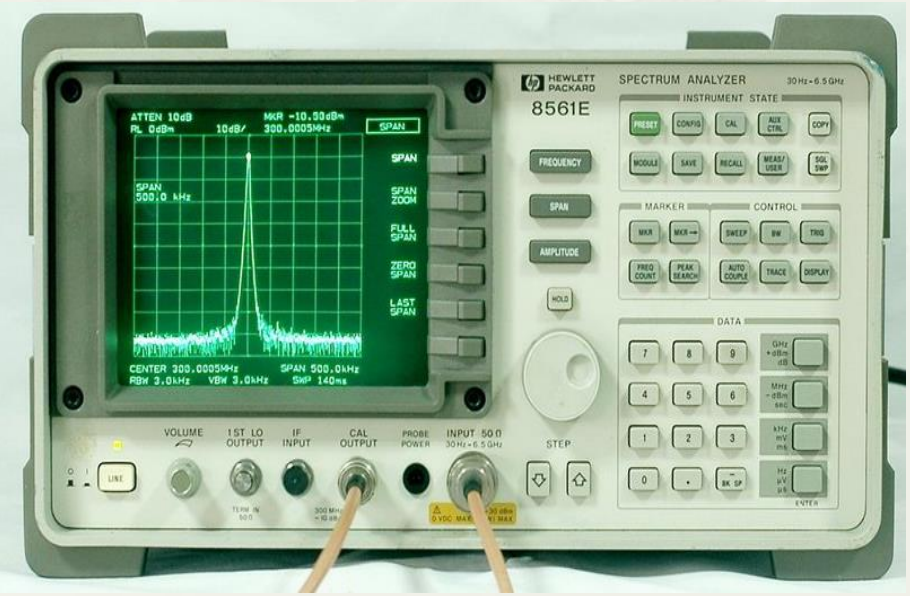
- Oscilloscope



Frequency-Domain Signals

- **Frequency-domain signals**
- **Spectrum**
  - The plot of amplitude versus frequency
  - Viewed by ***spectrum analyzer***

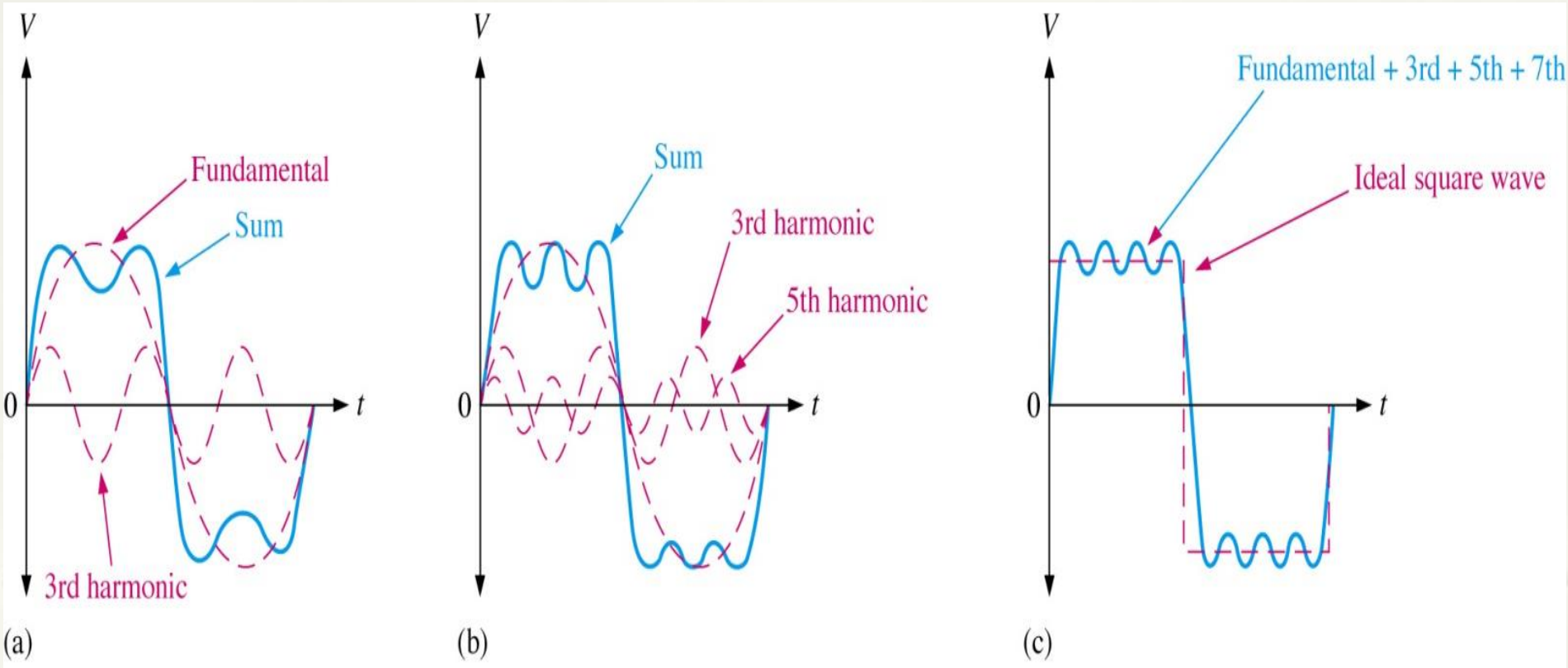




Harmonics

- **Fundamental frequency**
  - Basic repetition rate of the wave.
- **Harmonics**
  - Higher-frequency sinusoidal waves that are integer multiplies of the fundamental.
  - Odd harmonics – odd multiplies
  - Even harmonics – even multiplies

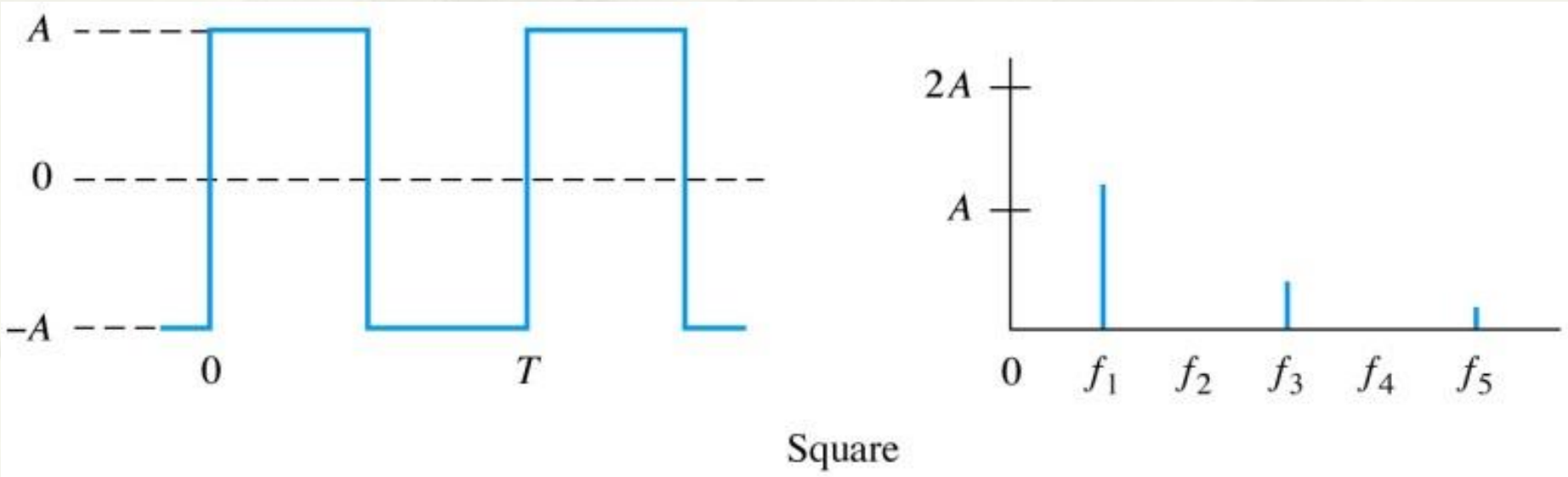
Harmonics



Odd harmonics combine to produce a square wave

Fourier Series

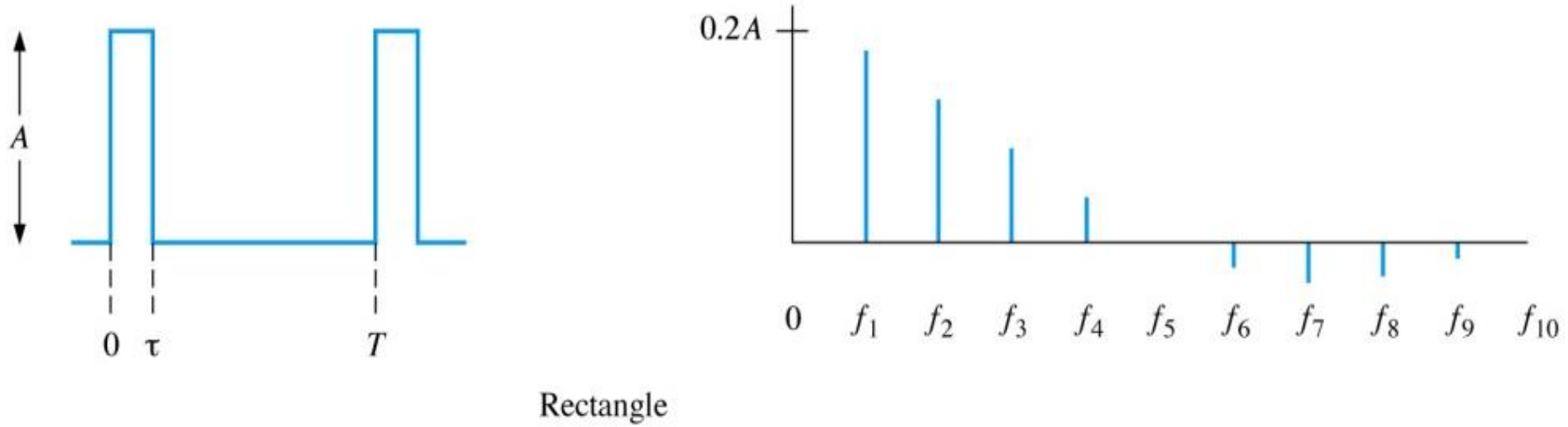
- **Fourier series & Fourier transform**
  - A mathematical series of trigonometry terms to describe periodic waves



Examples of time-domain and frequency-domain representations of repetitive waves

Fourier Series

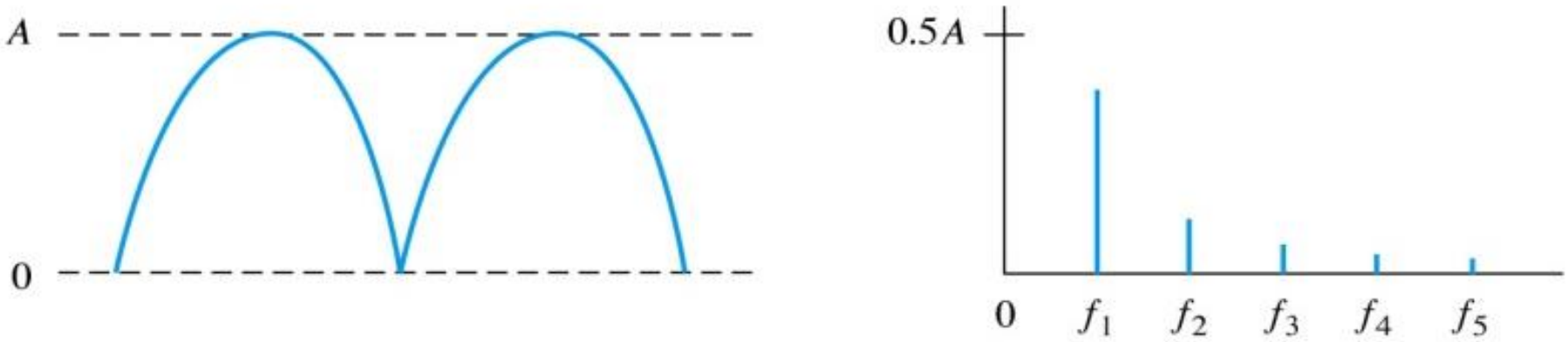
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Examples of time-domain and frequency-domain representations of repetitive waves

Fourier Series

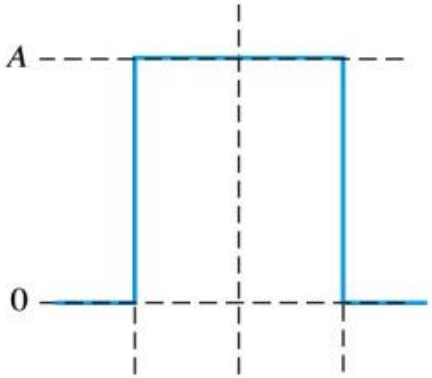
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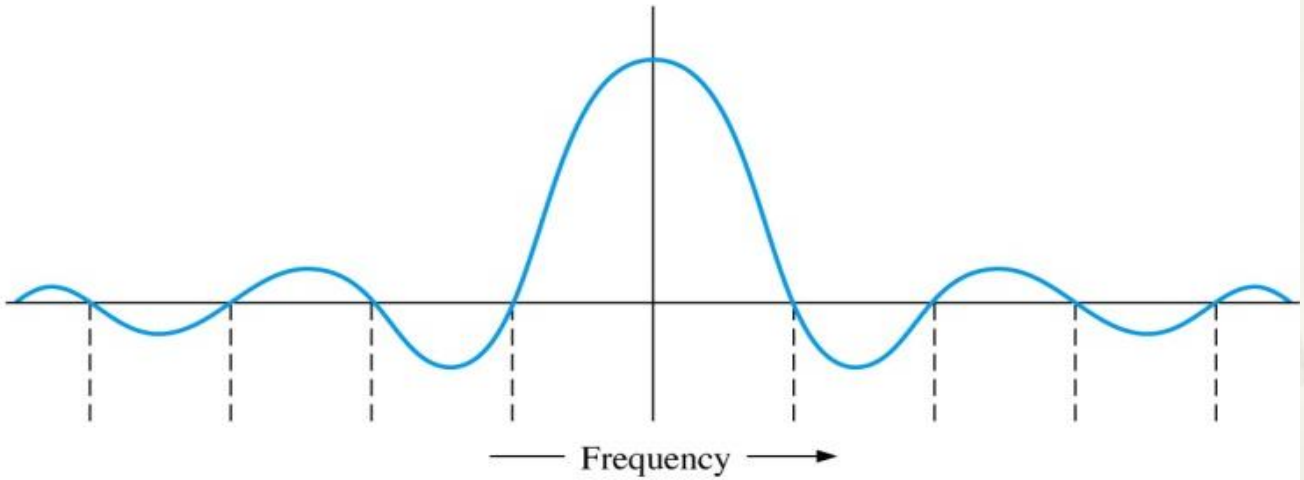
Examples of time-domain and frequency-domain representations of a nonrepetitive pulse waveform

Fourier Series

- **Fourier series & Fourier transform**
  - A mathematical series of trigonometry terms to describe periodic waves



Time-domain representation



Frequency-domain representation

Examples of time-domain and frequency-domain representations of a nonrepetitive pulse waveform

1-3 TROUBLESHOOTING ANALOG CIRCUITS

- Analysis, Planning, and Measuring
- Soldering
- Basic Test Equipment

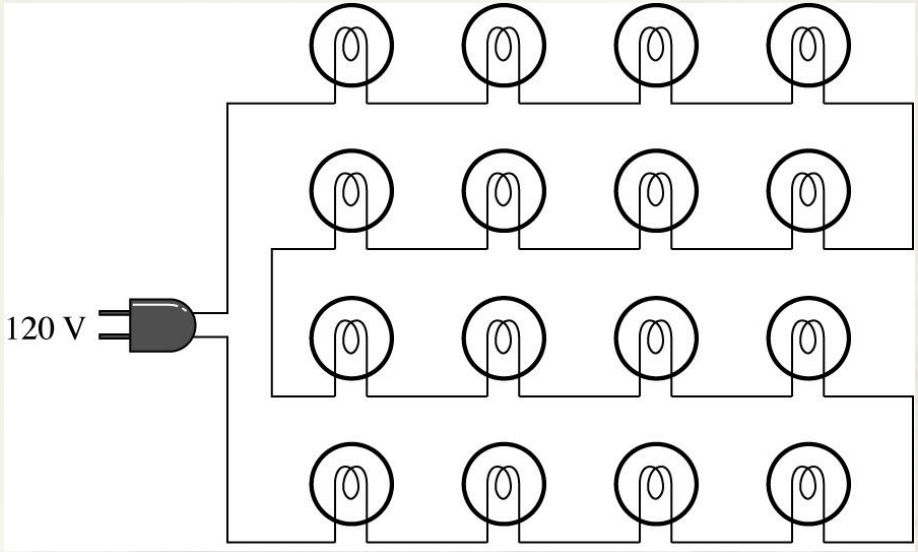
Troubleshooting

Diagnose and repair malfunctioning circuits and systems.

Analysis, Planning, and Measuring

- **Troubleshooting Procedure**
  1. Analyze the clues (symptoms) of the failure
  2. Form a logical plan
  3. Narrow the number of the possible failures by making carefully thought-out measurements

Troubleshooting Example



A series of lights

Half-splitting

- **A series of lights**
  - Suppose you have a string of 16 decorate lamps connected in series to a 120V source. Assume that this circuit worked at one time and stopped after moving it to a new location. When plugged in, the lamps fail to turn on.

Soldering

- **Soldering Tools and Materials**
  - Electric solder iron
  - Solder
  - Resin
  - Flux
  - Alcohol



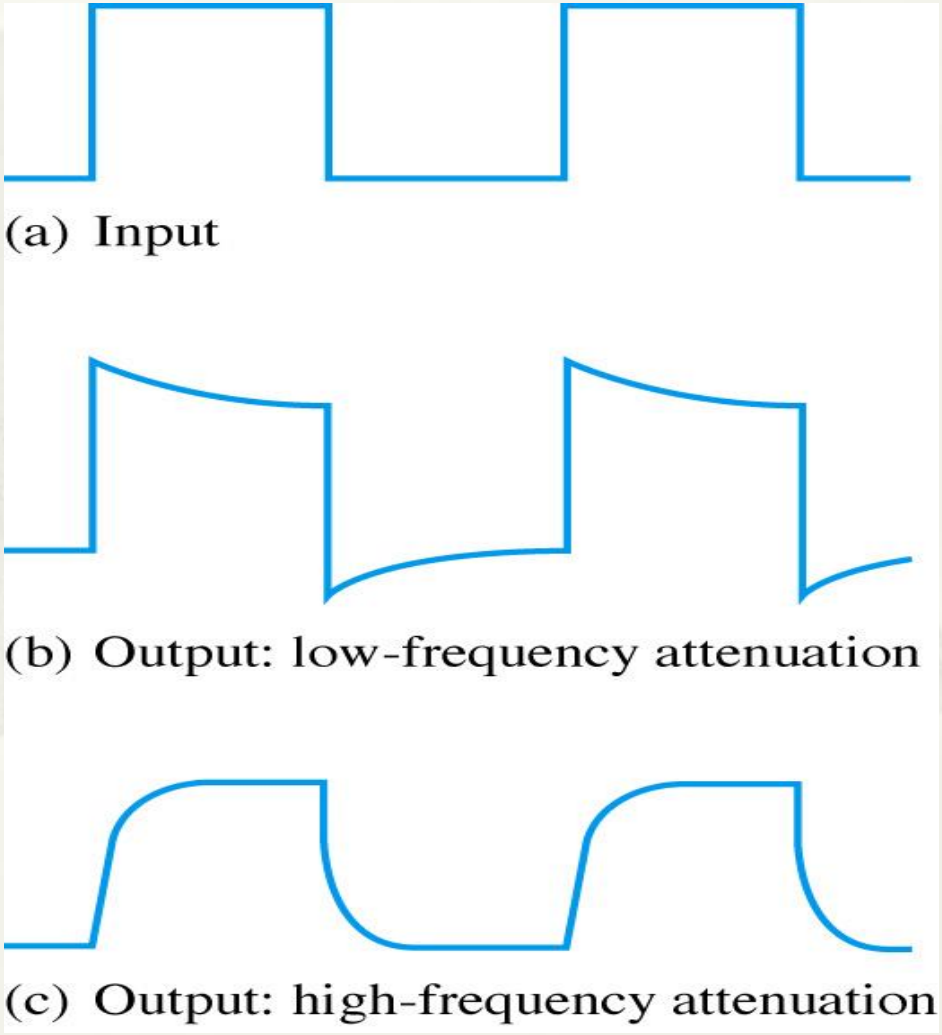
Soldering

- Remove the old part without damaging the board by excessive force or heat.
- Before installing a new part, the area must be clean.
- Don’t short together two parts or two pins on an integrated circuit by excessive solder.
- After the repair is completed, any flux must be removed from the board with alcohol or other cleaner.

Basic Test Equipment

- **Basic test equipments**
  - Oscilloscope
  - DMM (Digital Multimeter)
  - Power supply
  - VOM (Volt-Ohm-Milliammeter)
  - Function generator

Stimulus-response testing



Review Questions

1. What are two categories of integrated circuit?
2. What does the slope of an *IV* characteristic curve represent?
3. How does the *IV* characteristic curve of a large resistor compare to the *IV* characteristic curve of a small resistor?
4. What is the difference between dc and ac resistance?
5. What is the difference between conventional current and electron flow?

Review Questions

6. What is the difference between an analog signal and a digital signal?
7. What are the two steps required to convert an analog signal into a digital signal?
8. How is the average value of a sinusoidal wave calculated?
9. What is a harmonic?
10. How does the spectrum for a repetitive waveform differ from that of a nonrepetitive waveform?

Review Questions

11. What is the first step in troubleshooting a circuit?
12. What type of solder is used in electronics work?
13. What is meant by half-splitting?
14. What is meant by instrument loading?
15. How does testing with a square wave indicate an amplifier's response?